



Contents lists available at ScienceDirect

North American Journal of Economics and Finance



The relative size of exchange rate and interest rate responses to news: An empirical investigation

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ARTICLE INFO

Article history:

Received 25 March 2010

Received in revised form 20 May 2011

Accepted 5 July 2011

Available online 18 November 2011

Keywords:

Forward exchange rate

Events study

New Zealand

ABSTRACT

This paper examines the relative size of the effects of macroeconomic news on the spot exchange rate, and interest rate differentials (2- and 5-year swap rate differentials), and the synthetic forward exchange rate schedule, for the high-frequency New Zealand data. We find that the spot exchange rate and 5-year swap rates respond by a similar magnitude to monetary surprises, implying there is little response of the forward exchange rate to this type of news. In contrast, the spot exchange rate responds by nearly three times as much as 5-year interest rates to CPI and GDP surprises, implying that forward rates appreciate to higher than expected CPI or GDP news. This is in contrast to standard theoretical models and US evidence. Lastly, we show that exchange rates but not interest rates respond to current account news. The implications of these results for monetary policy are considered.

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1. Introduction

On October 26, 2006, the Reserve Bank of New Zealand left its policy rate, the Official Cash Rate (OCR), unchanged at 7.25 percent. This was a big surprise to the markets, which were expecting a 25 basis points of interest rate increase. The 30-day bank bill yield fell by 20 basis points immediately

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¹ The views expressed here are the views of the author and do not necessarily represent the views of the Reserve Bank of New Zealand.

after the announcement. The 1-, 2- and 5-year swap rates fell by 12, 12 and 9 basis points respectively in the 15 min window following the announcement, or by 1.62, 1.6 and 1.2 percent respectively. The New Zealand–US dollar exchange rate (NZD) fell by 0.92 percent in the same window or by a similar percentage amount as the 5-year swap rate. This episode is typical of the way the interest rates and the exchange rate simultaneously respond to monetary policy announcements in New Zealand.

In recent years, a large literature has examined how exchange rates and interest rates respond to macroeconomic news. This literature has typically shown that interest rates increase and exchange rates appreciate in response to unexpected monetary policy tightenings, stronger-than-expected news about the real economy, or positive inflation surprises. Yet, with the exception of Faust, Rogers, Wang, and Wright (2007), the relative size of the response of interest rates and exchange rates to news has scarcely been examined. This is surprising, not just because the relative sensitivity of different asset prices to news is informative about the strength of different monetary policy channels, but because the comparison can be interpreted as the response of the forward exchange rate to news.

In this paper we develop a simple framework to examine the relative responsiveness of exchange rates and interest rates to different types of news announcements. We show that the correlation between the change in the spot exchange rate and the change in different maturity interest rates is closely related to the way the whole spot-forward exchange rate schedule responds to news. In general, changes in the spot-forward exchange rate schedule can be characterized as having a shift component, measuring the change in the long term forward exchange rate, and a twist component, measuring the extent to which short term forward rates twist around the long term forward exchange rate because of changes in the term structure of interest rates.

This framework is used to examine how New Zealand's exchange rates and interest rates responded to four different types of macroeconomic news between 2000 and 2007. We find that the correlation patterns between the spot exchange rate and different maturity interest rates were different for different types of news, or, equivalently, that each type of news had a characteristic effect on the spot-forward exchange rate schedule. For example, we find that the exchange rate and long term (five years) interest rates changed by similar amounts in response to monetary policy announcements, or, equivalently, that the spot exchange rate twisted around an unchanging long term forward exchange rate pivot. In contrast, the spot exchange rate responded to current account announcements while interest rates did not, resulting in vertical shifts of the whole spot-forward exchange rate schedule. Most curiously, we find that the spot exchange rate responded by nearly three times as much as five-year interest rates to news about the inflation rate or real GDP. This means that higher than expected inflation and activity levels resulted in an appreciation of the five-year forward exchange rate plus an additional upward twist of the spot exchange rate. The inflation result is in contrast to standard economic models, in which long term forward exchange rates depreciate in response to higher than expected inflation, and is different than Faust et al.'s (2007) finding that the U.S. forward exchange rates depreciate in response to positive inflation news.

The paper introduces a small methodological twist to the literature. In the standard approach, which we largely follow, the change in an asset price during a small time period starting immediately before and ending soon after a news announcement is regressed against a numerical measure of the surprise in the announcement:

$$\Delta P_t^i = \alpha_0^{i,j} + \alpha_1^{i,j} W_t^j + u_t^{i,j} \quad (1)$$

where ΔP_t^i is the change in the i th asset and W_t^j is the measure of the j th surprise. The residual $u_t^{i,j}$ is the extent to which the price reaction differs from what can normally be expected given the surprise. The residual could be non-zero for various reasons. Of particular note, however, is the possibility that the markets respond to more detailed information than that which is observed by the econometrician and summarised in the numerical measure W_t^j . In this case we would expect the residual terms $u_t^{i,j}$ and $u_t^{k,j}$ for two different assets i and k to be correlated as the residuals reflect the response to the detailed information. In this paper we test for and exploit the correlation between these residual terms to refine tests of the hypothesis that the exchange rate and interest rate respond by the same amount to news. For each type of news we find that the slope coefficient between the two residual series has the same sign as the ratio $\alpha_1^{i,j}/\alpha_1^{k,j}$ and is similar in magnitude. Thus if interest rates respond by less

than the econometrician expects in response to a surprise, the spot rate also responds by less than expected.

The paper is organised as follows. In Section 2 there is a review of previous literature that has jointly estimated the response of exchange rates and interest rates to news, and the development of an alternative estimation strategy. In Section 3 we describe the data we use, and in Section 4 we present the results. Section 5 offers a discussion of the results and Section 6 concludes.

2. The estimation of the joint response of exchange rates and interest rates to news

2.1. Previous literature

In recent years a large literature has examined how exchange rates and interest rates respond to macroeconomic news. This “events analysis” literature stems back to the seminal piece by Kuttner (2001), which examined the effects on asset prices of changes in the Federal Funds Futures rate on the days of FOMC announcements. Kuttner assumed that the market prices prior to an announcement would incorporate all the available information up to that time and that the change in asset prices following the announcement would be due to the surprise only. Using this technique, the literature has shown that interest rates increase and exchange rates appreciate in response to unexpected monetary policy tightening, stronger than expected news about the real economy, or positive inflation surprises. Andersen, Bollerslev, Diebold, and Vega (2003) examine the response of the US dollar against Swiss franc, synthetic euro, British pound, Japanese yen and Deutsche mark for the period of 1992 and 1998. For a variety of macroeconomic announcements they find strong reaction by the exchange rate to economic news (or surprises). By using intra-day data Gürkaynak, Sack, and Swanson (2005) examines the effects of news on the interest rates. Subsequent work has examined the effects of monetary policy surprises on equity prices (Bernanke & Kuttner, 2005). Bernanke and Kuttner (2005) also considered how the impact of monetary policy surprises depends on the profile of future anticipated monetary policy. They showed that the effect of monetary policy announcements depended on the movement of the whole yield curve following an announcement, not just the change in short term rates. Related literature has considered the longer run impact of interest rates on the exchange rate, or the international transmission of these news on interest rates and exchange rates or equity prices (Craine & Martin, 2008; Ehrmann, Fratzscher, & Rigobon, 2011).

A small literature has already investigated how monetary and non-monetary surprises affect New Zealand interest rates and exchange rates. Zettelmeyer (2004) examined the response of exchange rates to interest rates using daily data. Kearns and Manners (2006) used high frequency data for Australia, New Zealand and the UK and showed monetary policy had a substantial influence on the exchange rate. They also showed that the impact of monetary policy surprises on the exchange rate depends on how the surprise affects expectations of future monetary policy. Karagedikli and Siklos (2008) examined the responses of the New Zealand dollar to monetary policy surprises and to the published interest rate forecast of the Reserve Bank of New Zealand. Drew and Karagedikli (2007) examined the effects of the New Zealand data surprises on the yield curve.

Although there have been a large number of studies investigating how macro events affect exchange rates or interest rates, to our knowledge Faust et al. (2007) is the only paper that has examined the simultaneous responses of interest rates and exchange rate to news. Faust et al. (2007) invoke the theory of uncovered interest parity to motivate their comparison of the way that exchange rates and interest rates respond to economic data releases. They note that to equalise returns on domestic and foreign interest earning assets, the expected exchange rate change over an n -year period must equal the n -year interest rate differential, adjusted for a risk premium:

$$Es_{t+n} - s_t = n(r_t^{f,n} - r_t^n) + n\rho_t^n \quad (2)$$

where $s_t = \ln(S_t)$, S_t is the spot exchange rate (foreign currency price of domestic currency), r_t^n and $r_t^{f,n}$ are domestic and foreign n -year interest rates, and ρ_n^t is the risk premium.¹ It follows that in a short interval surrounding the release of some economic news,

$$\Delta s_t = n(\Delta r_t^n - \Delta r_t^{f,n}) + \Delta Es_{t+n} - n\Delta \rho_n^t \quad (3)$$

Faust et al. (2007) use this equation to analyse the behaviour of ΔEs_{t+n} in response to news under the assumption that $\Delta \rho_{t,n} = 0$, and to analyse the behaviour of $\Delta \rho_{t,n}$ in response to news under the assumption ΔEs_{t+n} follows a random walk.

As the synthetic n -year forward rate F_t^n is defined as

$$F_t^n = S_t \left(\frac{1 + r_t^{n,f}}{1 + r_t^n} \right)^n \quad (4)$$

with

$$\Delta f_t^n \simeq \Delta s_t + n(\Delta r_t^{n,f} - \Delta r_t^n) \quad (5)$$

where $f_t^n = \ln(F_t^n)$, the analysis of ΔEs_{t+n} under the assumption that $\Delta \rho_{t,n} = 0$ is the same as the analysis of Δf_t^n . From Eq. (5) it is apparent that the extent that the spot exchange rate responds to news more than the n -year interest rate differential is identical to the response of the n -year forward rate to news. By examining the response of the whole spot-forward exchange rate schedule to news, the relative size of spot exchange rate changes and different maturity interest rate changes can be readily ascertained.

In general, changes in the spot-forward exchange rate schedule can be characterized as having a shift component, measuring the change in the long term forward exchange rate, and a twist component, measuring the extent to which short term forward rates twist around the long term forward exchange rate because of changes in the term structure of interest rates. This can be seen by rearranging Eq. (5) as a function of the interest rate differential and the longest maturity forward exchange rate, f_t^L :

$$\Delta f_t^n \simeq \Delta f_t^L + (L\Delta r_t^L - n\Delta r_t^n) - (L\Delta r_t^{L,f} - n\Delta r_t^{n,f}) \quad (6)$$

The second term on the right hand side captures the extent that the total return on a long maturity bond increases in response to news by more than the total return on a short dated (n -year) bond. If this term is positive, it means that short term interest rates are expected to increase between periods n and L . Consequently, the extent to which short term forward exchange rates increase with respect to the long term forward rate reflects both the size of the change in short term interest rates and the total length of time that the news is expected to affect short term interest rates in the future.²

Eq. (6) can be used to interpret recent work by Clarida and Waldman (2007), who developed a simple rational expectations model to investigate their empirical finding that spot exchange rates often appreciate in response to higher than expected inflation news. Their argument is that the spot exchange rate should appreciate if the central bank is expected to increase interest rates to offset inflationary pressure, even though in the long run the exchange rate should depreciate to ensure purchasing power parity holds. In the language we use, their model posits that there should be an appreciation of the spot rate but a depreciation of the long term forward rate. Equivalently, by Eq. (6), their theory predicts that the spot exchange rate appreciates in response to positive inflation surprises, but it does so by less than the increase in the total return to a long term bond. Clarida

¹ Note we have changed their nomenclature to be consistent with New Zealand usage as New Zealand traditionally quotes exchange rates as the foreign price of the local currency. In countries such as the United States where exchange rates are quoted as the local price as the foreign currency, Eq. (1) is expressed as $Es_{t+n} - s_t = n(r_t^n - r_t^{f,n}) + n\rho_n^t$.

² An example makes this clear. Suppose in response to news the one-year interest rate was expected to increase by 1 percent for three years, but there was no change to the one-year rate thereafter. The total return to a one-year bond would increase by 1 percent, the total return to a two-year bond would increase by 2 percent, and the total return to all bonds with maturities of three years or more would increase by 3 percent. Thus the spot rate would appreciate 3 percent with respect to the long forward rate, the 1-year forward rate would appreciate by 2 percent with respect to the long forward rate, but there would be no change in the relative value of the 3-year forward rate with respect to the long forward rate.

and Waldman examine the behaviour of spot exchange rates but do not test their implicit hypothesis about the behaviour of the long term forward rate. Below we show that spot and forward exchange rates both appreciate in response to positive inflation surprises in New Zealand, in contrast to their hypothesis.

There has been a recent and influential literature that has emphasized the role of monetary policy in determining how the exchange rate responds to changing expectations about the future value of economic fundamentals. Both Engel and West (2005) and Engel, Mark, and West (2007) use a Taylor rule model to consider how exchange rates change when there is an endogenous response of monetary policy to economic news. Consistent with Faust et al. (2007), their theoretical models show “bad news” about inflation should be “good news” for the exchange rate. They show that if the monetary authority increases short term interest rates in response to factors that raise the inflation rate above the target level, and if uncovered interest rate parity holds, the spot exchange rate will appreciate not depreciate in response to economic fundamentals that increase inflationary pressure. Engel et al. (2007) further argue that empirical evidence, including evidence from event studies, about the behaviour of the exchange rate in response to inflation and the output gap is consistent with these models. The evidence they compile suggests that an expanding economy (or higher than expected activity) or higher inflation in the home country relative to other countries leads to an appreciation of the currency.

The analysis in this paper is consistent with these theoretical models. Their Taylor rule model shows how the spot exchange rate responds to inflation and the output gap. This paper analyses how the spot and forward exchange rates respond to these variables. In Appendix A, we extend the Engel et al. (2007) Taylor rule model by calculating the effect of inflation, the output gap, and risk premia on both spot and forward exchange rates. We show that while in their model the spot exchange rate should appreciate in response to higher than expected inflation, the long term forward rate should depreciate. The effect of higher than expected output on the long term forward rate is ambiguous however. Although higher than expected output should make the spot rate appreciate, it should only lead to an appreciation of the forward exchange rate if local prices are expected to fall in the long run.

2.2. Estimation strategy

In this paper we focus on the joint response of the spot exchange rate and the five-year swap rate differential to news, although we also examine shorter horizons. We assume the relationships between the spot exchange rate, interest rate differentials, and news announcements can be described by the following three equations:

$$\Delta s_t = \alpha_0^{s,j} + \alpha_1^{s,j} W_t^j + u_t^{s,j} \quad (7)$$

$$n(\Delta r_t^n - \Delta r_t^{n,f}) = \alpha_0^{rn,j} + \alpha_1^{rn,j} W_t^j + u_t^{rn,j} \quad (8)$$

$$u_t^{s,j} = \delta^{n,j} u_t^{rn,j} + \eta_t^n \quad (9)$$

The covariance matrix is

$$\Sigma^{nj} = \begin{pmatrix} \sigma_{sj}^2 & \sigma_{smj} \\ \sigma_{smj} & \sigma_{rnj}^2 \end{pmatrix}$$

and $\delta^{n,j} = \sigma_{smj} / \sigma_{rnj}^2$.

To simplify the following exposition, we drop the superscripts or subscripts j and n , rename $\Delta R_t = n(\Delta r_t - \Delta r_t^f)$, ignore the constant terms and write the system in matrix form:

$$\Delta s = W\alpha^s + u^s \quad (10)$$

$$\Delta R = W\alpha^r + u^r \quad (11)$$

$$u^s = u^r \delta + \eta \quad (12)$$

The coefficients α^s and α^r can be efficiently estimated using ordinary least squares. Let $\hat{\alpha}^s$ and $\hat{\alpha}^r$ be the OLS estimated coefficients of α^s and α^r in Eqs. (10) and (11), $\hat{u}^s$ and $\hat{u}^r$ the corresponding residual estimates, $\hat{\Sigma}$ the estimate of the covariance matrix made using these residuals, and $\hat{\delta}$ the estimate of the coefficient δ in Eq. (12).

There are two ways to test whether the spot exchange rate and the interest rate differential respond to news by the same amount, or equivalently whether the forward rate responds to news. First, one can calculate the forward rate $\Delta f = \Delta s - \Delta R$, and estimate the equation

$$\Delta f = W\alpha_f + u_f \quad (13)$$

It follows from (10) and (11) that $\alpha_f = \alpha_s - \alpha_r$. A simple test of the hypothesis that $\alpha_s = \alpha_r$ is therefore to estimate (13) using OLS and test whether $\alpha_f = 0$. Note that the variance of u_f is $\sigma_f^2 = \sigma_s^2 + \sigma_r^2 - 2\sigma_{sr}$, the variance of the OLS estimator of α_f is $\sigma_{\hat{\alpha}_f}^2 = \sigma_f^2(W'W)^{-1}$ and the OLS estimate of α_f is $\hat{\alpha}_f = \hat{\alpha}^s - \hat{\alpha}^r$.

The second approach is to decompose the change in the interest rate differential ΔR into a predictable component $\Delta R^* = W\alpha_r$ and an unpredictable component u_r , and then to regress the spot rate against both:

$$\Delta s = \Delta R^* \gamma_0 + u_r \gamma_1 + v \quad (14)$$

From (10), (11), and (12),

$$\Delta s = \Delta R^* \frac{\alpha_s}{\alpha_r} + u_r \delta + \eta \quad (15)$$

$$\Delta f = \Delta R^* \left(\frac{\alpha_s}{\alpha_r} - 1 \right) + u_r(\delta - 1) + \eta \quad (16)$$

Thus Eq. (13) is a version of Eq. (16) with δ restricted to equal 1.

If we knew the true values of ΔR^* and u_r , we could estimate Eq. (14) directly and test the hypothesis $\alpha_s = \alpha_r$ by testing whether $\gamma_0 = 1$. The advantage of this approach over the first approach is that the standard error of the coefficient γ_0 is smaller than the standard error of α_f unless $\delta = 1$: this is because $\sigma_{\hat{\eta}}^2 \leq \sigma_f^2$.³

In practice, of course, we do not know ΔR^* and u_r . Rather, we use the OLS estimates of Eq. (11) to generate estimates $\Delta \hat{R}^*$ and $\hat{u}_r$, and we estimate the generated regression

$$\Delta s = \Delta \hat{R}^* \theta_0 + \hat{u}_r \theta_1 + \xi \quad (17)$$

This is an example of a type of equation examined by Pagan (1984). The OLS estimates are $\hat{\theta}_0 = \hat{\alpha}_s / \hat{\alpha}_r$, $\hat{\theta}_1 = \hat{\delta}$, and $\hat{\xi} = \hat{\eta}$. Hence Eq. (17) can be used to test the hypothesis that $\alpha_r = \alpha_s$ by testing the hypothesis $\theta_0 = 1$. Note, however, that it is no longer the case that Eq. (17) automatically produces a better test of the hypothesis than Eq. (13). Pagan showed the variance of the OLS estimate of θ_0 is $\sigma_{\hat{\theta}_0}^2 = (\sigma_{\hat{\eta}}^2 + \sigma_r^2(\theta_0 - \theta_1)^2)(W'W)^{-1}$.⁴ This is smaller than the variance of the OLS estimate of α_f if $\sigma_r^2(\theta_0 - \theta_1)^2 + \sigma_{\hat{\eta}}^2 \leq \sigma_f^2$: since $\sigma_f^2 = \sigma_s^2 + \sigma_r^2(1 - \delta)^2$, this condition is equivalent to $(\theta_0 - \theta_1)^2 \leq (1 - \delta)^2$. Thus if δ is closer to the ratio α_s / α_r than to 1, Eq. (17) is likely to produce a more accurate test of the hypothesis that $\alpha_r = \alpha_s$ than Eq. (13). As we show below, this condition is likely to be satisfied for the CPI and GDP surprises, where we estimate both the ratio α_r / α_s and the coefficient δ to be greater than 2. In contrast, because current account news has very little effect on long term interest rates, the estimates of α_r and δ for the current account surprises are very near zero and Eq. (13) provides a much better test of the hypothesis than Eq. (17). For monetary policy surprises we estimate that δ is very close to 1 and thus Eq. (13) provides an efficient method of testing the hypothesis.

The above analysis emphasizes the role the parameter δ plays in the efficiency of tests that examine how exchange rates and interest rate differentials respond to news. We think there are good grounds to believe that this parameter will be neither 0 nor 1 in general. One reason is that residual terms u_s

³ $\sigma_f^2 = \sigma_s^2 + \sigma_r^2 - 2\sigma_{sr}$, $\sigma_{\hat{\eta}}^2 = \sigma_s^2 + \delta^2\sigma_r^2 - 2\delta\sigma_{sr}$. Hence $\sigma_f^2 - \sigma_{\hat{\eta}}^2 = \sigma_r^2(1 - \delta)^2 \geq 0$.

⁴ He further showed that the OLS estimate of $\sigma_{\hat{\theta}_0}^2$ is inconsistent. Corrected estimates are presented in the table below.

and u_r in Eqs. (10) and (11) may contain the response of market participants to detailed information that is not captured by the summary measure W . A 0.2 percent higher than expected increase in the CPI that is attributed to rising oil prices may elicit a very different financial market response than a similar sized increase in the CPI that is associated with an across the board rise in non-tradeable prices, for example, and the response is likely to be correlated across different financial markets. If this is the main reason why u_s and u_r are correlated, and if market participants respond to the detailed information in the same way that they respond to the summary information W , the coefficient δ will tend to the ratio α_s/α_r .

Lastly we estimate the correlation between the spot exchange rate changes and the interest rate differential changes over the news announcement periods:

$$\Delta s = \Delta R\beta + \varepsilon \quad (18)$$

This equation is a version of Eq. (13) with the coefficients γ_0 and γ_1 restricted to be equal.

3. Data

In Section 4 we estimate how the interest rate, spot exchange rate and forward exchange rates respond to new information in data and monetary policy announcements. The exchange rate response is measured as the change in the logarithm of the exchange rate over a period starting immediately before the announcement and ending shortly after the announcement. The implicit forward rates were calculated from the spot exchange rate and the 2- and 5-year interest swap rates of the New Zealand dollar and the U.S. dollar. Drew and Karagedikli (2007) show that swap rates with horizons longer than five years behave in almost exactly the same manner as five-year swap rates so we limit our analysis to these maturities. The intra-day swap and spot exchange rates were obtained from the Marketwatch Databank of the Reserve Bank of New Zealand. We have data on 2- and 5-year interest rate swaps at 10 min past every hour. The spot exchange rate data are collected at the same time.⁵

We use the market-based measure of monetary surprises used; basic measure of a monetary surprise is the change in the 90-day bank bill yields over the period immediately surrounding the policy change announcement. Thus if the 90-day bill rate fell from 8.40 to 8.25 percent after the Governor announced there would be no change in the overnight cash rate (OCR), the interest rate instrument the Reserve Bank of New Zealand uses, we would measure the surprise as -0.15 percent.⁶

This measure of monetary surprises is similar in spirit to the ones used in the literature: for example, Kuttner (2001) and Bernanke and Kuttner (2005) calculate a measure of U.S. monetary surprises from the Fed funds futures rates. In addition to monetary surprises, we analyse the effect of three data announcements: Gross Domestic Product (GDP), the Consumer Price Index (CPI) and the Current account (CA). The surprise components of these data announcements is measured as the difference between the median market expectations surveyed by Bloomberg and the actual out-turns. This measure of a surprise is now standard and believed to capture changes in expectations quite accurately (see Engel, 2007).

We considered using data windows of three different lengths: one hour (the tight window), two hours (the wide window) and one day (the daily window). The choice of the length of windows is dictated by the availability of data. The data we have is hourly (recorded 10 min past every hour). Table 1 shows the data windows we used. Monetary policy announcements are made at 9.00 am in New Zealand. The last reading of interest rates and exchange rates available to us were at 8.10 am, and the first reading after the announcements were at 9.10 am. The tight window is defined as the log difference in the market rates between 8.10 am and 9.10 am, and thus measures the market reaction after 10 min. There is an asymmetry between the time before and after the announcement in this

⁵ Tick by tick data are available for all major New Zealand variables but not for foreign swap rates. Consequently, we opted for hourly data.

⁶ See Drew and Karagedikli (2007) and Karagedikli and Siklos (2008) for more details. Our results are similar to the spot exchange rate results of the former and the interest rate results of the latter, with some minor differences arising from the different data windows. Drew and Karagedikli (2007) and Karagedikli and Siklos (2008) also use two other measures of monetary surprises in their papers, finding results that are very similar to those based on the market-based measure.

Table 1
Data windows.

Window	Monetary surprise	Data surprise
Announcement	9.00 am	10.45 am
Tight window	8.10 am to 9.10 am	10.10 am to 11.10 am
Wide window	8.10 am to 10.10 am	10.10 pm to 12.10 am
Daily window	8.10 pm to 17.10 am	10.10 pm to 17.10 am

window, hence we consider two other windows: “Wide window”, which is the log difference of the rate between 10.10 am and 8.10 am and “daily window”, which is the log difference of the rate between 17.10 pm and 8.10 am. The wide window is our preferred window as the tight window only measures the market reaction ten minutes after the announcement.

In New Zealand the GDP, CPI and current account data are announced quarterly at 10.45 am. The tight window is the log difference between the 11.10 am and the 10.10 readings. The tight window measures the reaction of the market 25 min after the announcement. This is a long enough time, and this is our preferred window for the data announcements. In addition, we also estimate the reactions with two other windows: the wide and the daily windows, which are the log differences of the market rates between 12.10 pm and 10.10 am, and 17.10 pm and 10.10 am respectively. Our preference is to use a shortest period window possible, for the standard errors of the coefficients in longer period windows are larger as there is more scope for the exchange rate and interest rate to change in response to other factors. This is also found in *Gürkaynak et al. (2005)*. So, throughout the text we refer to the wide window in the case of monetary policy surprises, and to the tight window in the case of data surprises. We also report the results from the other windows.⁷

4. Results

Tables 2–5 show the results of the estimation of Eqs. (10)–(18) using the four different sets of macroeconomic surprises. The top half of each table displays the results using two-year interest rate swap rates and forward rates, while the bottom half displays the results using five-year rates. The first row of each table shows the *total* response of the two-year interest rate differential to a surprise, $\Delta R_2 = 2(\Delta r^{2,NZ} - \Delta r^{2,US})$. This can be interpreted as the additional return over the life of the swap contract that is earned following a surprise, and thus can be directly compared to the percentage change in the spot exchange rate. We focus on the five-year results on a priori grounds as the two-year forward rate is unlikely to be a good proxy for the long term forward rate.

Because the regressions are linear one variable equations, they can easily be shown graphically. Figs. 1–3 show how the 5-year swap rate, the spot exchange rate, and the 5-year forward exchange rate responds to monetary policy announcement surprises (these graphs corresponds to the second half of Table 2). It is clear from Figs. 1 and 3 that positive monetary policy surprises cause the 5-year swap rates and the spot exchange rate to appreciate. The difference in the slopes of these graphs is the relationship between the forward rate and the surprises, shown in Fig. 3. While positive, the slope is much smaller and is not different than zero at conventional significance levels.

More formally the results in Table 2 show that a 1 percentage point surprise increase in the OCR leads to a 2.4 percentage point increase in the total earnings of a five-year bond.⁸ The exchange rate appreciates by 3.6 percent, or by 1.2 percentage points more than the increase that might be expected given the total increase in the return from a five-year bond. Equivalently, this means the five-year forward rate appreciates by 1.2 percent in response to a 1 percentage point OCR surprise. The difference is not statistically significant at the 5 percent level, however, indicating that the spot exchange rate responds in a similar fashion to OCR news as five-year bonds. Put differently, the spot

⁷ Unfortunately the US interest rates are the only data that is recorded at 10 min past every hour, which is the main reason why we use this kind of hourly data.

⁸ Drew and Karagedikli indicate that the total earnings increase in longer maturity bonds is similar i.e. it is also 2.4 percentage points.

Table 2

NZ–US dollar response to monetary policy surprises.

Dependent variable	W	$\hat{u}_r$	$\Delta \hat{R}$	ΔR	Obs	R ²	t-Test $\alpha_5 = \alpha_r$
2-Year rates							
ΔR_2	0.013** (0.0019)				55	0.48	
Δs	0.036** (0.0081)				55	0.22	
Δf_2	0.0226* (0.0087)				55	0.22	2.60*
$\hat{u}_5$		1.81** (0.63)			55	0.14	
Δs		1.81** (0.63)	2.70** (0.66)		55	0.33	2.57*
Δs				2.24** (0.45)	55	0.32	
5-Year rates							
ΔR_5	0.024** (0.0035)				55	0.46	
Δs	0.036** (0.0091)				55	0.22	
Δf_5	0.012 (0.0084)				55	0.038	1.44
$\hat{u}_5$		1.06** (0.33)			55	0.16	
Δs		1.06** (0.33)	1.51** (0.36)		55	0.35	1.41
Δs				1.27** (0.24)	55	0.34	

* Statistical significance at 5% level.

** Statistical significance at 1% level.

exchange rate largely twists around the forward pivot in response to news about monetary policy, although there is a modest (and statistically insignificant) shift of the long forward rate to this news.

The response of interest rates and exchange rates to CPI and GDP news is quite different. In each case, the spot exchange rate is nearly three times more responsive to news than the five-year swap rate differentials. For example, total earnings on a five-year swap rate contract increase by 0.32 percentage points in response to a 1 percentage point surprise increase in the CPI, whereas the exchange rate

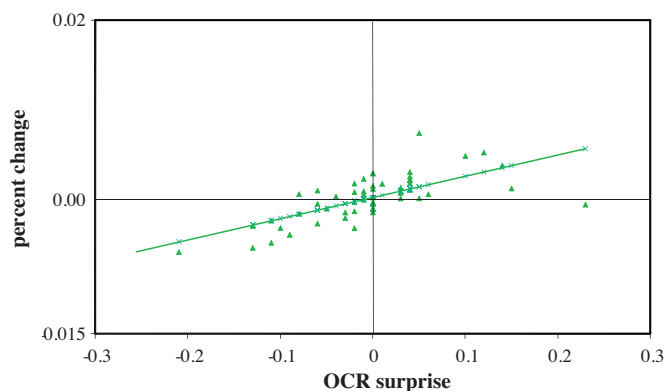
**Fig. 1.** US–NZ interest differential movements in response to OCR surprises.

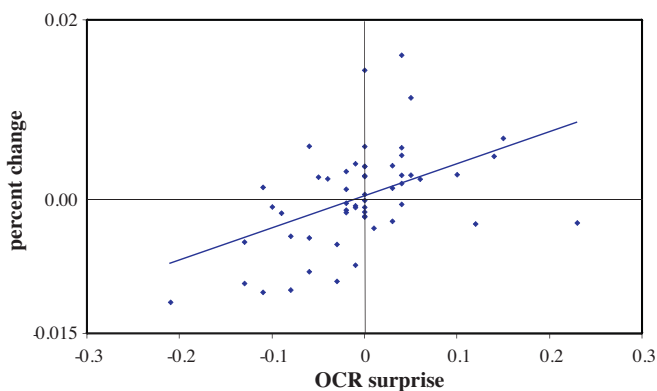
Table 3

NZ–US dollar response to CPI surprises.

Dependent variable	W	$\hat{u}_t$	$\Delta \hat{R}$	ΔR	Obs	R^2	t-Test $\alpha_s = \alpha_f$
2-Year rates							
ΔR_2	0.0024** (0.00057)				23	0.45	
Δs	0.0087** (0.0024)				23	0.37	
Δf_2	0.0063* (0.0024)				23	0.24	2.59*
$\hat{u}_s$		0.89 (0.94)			23	0.04	
Δs		0.89 (0.94)	3.65** (1.25)		23	0.40	2.12*
Δs				2.12** (0.74)	23	0.28	
5-Year rates							
ΔR_5	0.0032** (0.00057)				23	0.59	
Δs	0.0087** (0.0024)				23	0.37	
Δf_5	0.0055* (0.0022)				23	0.23	2.49*
$\hat{u}_s$		2.41** (0.78)			23	0.32	
Δs		2.41** (0.78)	2.71** (0.66)		23	0.57	2.61*
Δs				2.59** (0.49)	23	0.57	

* Statistical significance at 5% level.

** Statistical significance at 1% level.

**Fig. 2.** US dollar spot-rate movements in response to OCR surprises.

appreciates by 0.87 percent.⁹ The differences between the exchange rate and interest rate differential response to CPI and GDP surprises are quite accurately measured and are statistically significant at the five percent significance level. Thus it can be concluded that in response to higher than expected CPI and GDP news there is an upward shift of the forward rate and a further upward twist of the spot rates that reflects higher local interest rates.

⁹ Note the typical surprise in the CPI is only 0.125 percent.

Table 4

NZ–US dollar response to GDP surprises.

Dependent variable	W	$\hat{u}_t$	$\Delta \hat{R}$	ΔR	Obs	R ²	t-Test $\alpha_5 = \alpha_r$
2-Year rates							
ΔR_2	0.0016** (0.00047)				23	0.35	
Δs	0.0064** (0.0019)				23	0.34	
Δf_2	0.0048* (0.0019)				23	0.24	2.57*
$\hat{u}_5$		1.39 (0.86)			23	0.12	
Δs		1.39 (0.86)	3.93* (1.39)		23	0.41	2.10*
Δs				2.29** (0.72)	23	0.32	
5-Year rates							
ΔR_5	0.0022** (0.00054)				23	0.43	
Δs	0.0064** (0.0019)				23	0.34	
Δf_5	0.0042* (0.0016)				23	0.24	2.54*
$\hat{u}_5$		2.35* (0.59)			23	0.44	
Δs		2.35** (0.59)	2.89** (0.68)		23	0.63	2.73*
Δs				2.58** (0.44)	23	0.62	

* Statistical significance at 5% level.

** Statistical significance at 1% level.

Although the spot exchange rate responds more than the interest rate differential to GDP and CPI news, the size of the exchange rate response is not very large. A one percentage point increase in GDP or the inflation rate is associated with a 0.64 or 0.87 percent appreciation of the spot exchange rate respectively. These movements, while non-trivial, suggest that the normal variation in GDP and the inflation rate is only responsible for a small amount of the variation in the currency.

These results confirm the finding by [Clarida and Waldman \(2007\)](#) that bad news about the inflation rate is good news about the spot exchange rate. In their theoretical model, however, they imply that

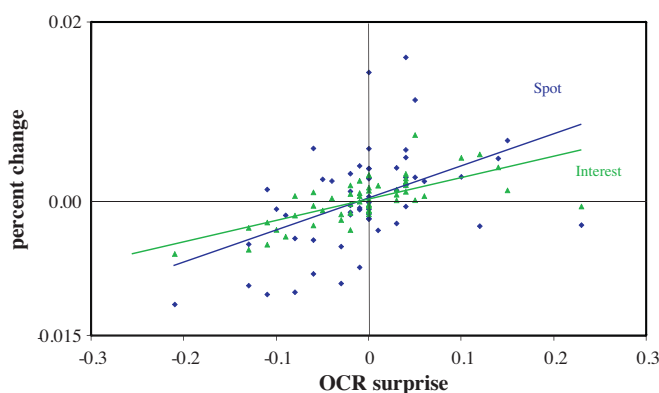
**Fig. 3.** US dollar and US–NZ interest rate differentials in response to OCR surprises.

Table 5

NZ–US dollar response to current account surprises.

Dependent variable	W	$\hat{u}_r$	$\Delta \hat{R}$	ΔR	Obs	R^2	t -Test $\alpha_s = \alpha_r$
2-Year rates							
ΔR_2	–0.00017 (–0.0002)				22	0.04	
Δs	0.0044** (0.0007)				22	0.65	
Δf_2	0.0046** (0.0007)				22	0.68	6.20**
$\hat{u}_s$		0.95 (0.80)			22	0.07	
Δs			–25.6 (32.0)		22	0.67	0.83
Δs				0.10 (1.3)	22	0.00	
5-Year rates							
ΔR_5	0.000 (0.0003)				22	0.00	
Δs	0.0044** (0.0007)				22	0.65	
Δf_5	0.0044** (0.0081)				22	0.60	5.50**
$\hat{u}_s$		–0.22 (0.60)			22	0.01	
Δs		–0.22 (0.60)	5129 (1.6×10^6)		22	0.65	0.003
Δs				–0.22 (0.99)	22	0.00	

** Statistical significance at 1% level.

the long term forward rate should depreciate in response to higher than expected CPI news, rather than appreciate. Faust et al. (2007) also find that US forward rates with respect to the Euro and sterling depreciate in response to stronger than expected CPI news. Consequently, the evidence presented here that the New Zealand long term forward exchange rate appreciates in response to CPI surprises – that the spot rate responds much more than swap rates to CPI news – is unusual. It is possibly related to the evidence presented by Clarida and Waldman (2007) that the New Zealand dollar spot exchange rate is much more responsive to CPI news than any other developed country currency other than the Norwegian krone.

In each of these three cases, the two-year bond rate changes less in response to news than the five-year bond rate. This implies that news is expected to have an effect on short term interest rates between two and five years into the future, a finding consistent with the evidence presented in Drew and Karagedikli (2007). As a result, the two-year forward exchange rate appreciates in response to positive economic news by more than the five-year forward exchange rate, but by less than the spot exchange rate. It appears, therefore, that positive economic news causes an upward twist of the spot-forward exchange rate schedule around the long term forward rate.

The response of interest rates and exchange rates to current account news is different again. Table 5 shows that interest rates are unaffected by current account news – the coefficients on the interest rate terms are small and statistically insignificant – whereas the spot rate appreciates in response to a positive current account news. As a consequence, spot and forward exchange rates increase by the same amount in response to macroeconomic news.

The regressions reveal that the residuals u_r and u_s are strongly positively correlated in the equations for monetary policy, CPI and GDP surprises, with the spot estimates of the coefficients δ_5 equal to 1.06, 2.41 and 2.35 respectively. (Since there is no relationship between current account surprises and interest rates, the correlation coefficient is insignificantly different to zero.) In each case the coefficients are of similar magnitude to the ratio α_s/α_r . These results are consistent with the idea that the markets

respond to a more detailed level of information than that represented by the headline figure, and that the response to this detailed information is consistent with the response to the headline figure. Consequently, an econometrician can predict how the exchange rate responds to news much better if he or she knows both the headline news announcement and the response of interest rates to this news rather than just the headline announcement. Incorporating the interest rate variable as well as the news variable into the spot equation regression increased the R^2 term from 0.22 to 0.35 for OCR surprises, 0.37 to 0.57 for CPI surprises, and 0.34 to 0.63 for GDP surprises.

The small size of the sample we use means that the coefficients δ and the ratio α_s/α_r could not be estimated with sufficient accuracy. According to the theory, in the limit Eq. (17) would have better efficiency than Eq. (13) so long as δ is closer to α_s/α_r than 1. For monetary policy surprises, $\hat{\delta} = 1.06$ so this is unlikely to be the case. For CPI and GDP surprises, $(\hat{\delta}, \hat{\alpha}_s/\hat{\alpha}_r) = (2.41, 2.71)$ and $(2.35, 2.89)$ respectively, so some improvement is likely. For current account surprises, Eq. (17) has very poor properties because when interest rates are uncorrelated with news the ratio $\alpha_s/\alpha_r \rightarrow \infty$. The results of the two t -tests of the hypothesis $\alpha_s = \alpha_r$ were similar for the OCR, CPI, and GDP surprises, but not for current account surprises. In the latter case, the test associated with Eq. (13) decisively rejected the hypothesis although the test associated with Eq. (17) did not.¹⁰

5. Discussion

Central banks around the world routinely focus on spot exchange rates rather than forward exchange rates. The usual reason for the neglect of forward exchange rates is that they tend to be residually calculated from spot exchange rates and bond rates and thus contain little independent information. Obviously it is true that a forward exchange rate can be calculated from a spot exchange rate and a local-foreign interest rate differential. Nonetheless, it does not mean that changes in the forward rate are devoid of informational content, for they provide information about the correlation of movements in the spot exchange rate and the interest rate differential. This correlation can be calculated and presented directly, or it can be presented in the guise of the way the forward exchange rate behaves. Sometimes it may be more informative to present in terms of the co-movements of spot exchange rates and interest rates. On other occasions, it may be useful to analyse the forward rate directly.

In the short period we studied there were three interesting results. First, the spot exchange rate and the five-year swap rate differential have a similar response to monetary policy surprises. This means the medium and long term forward rates change little in response to monetary policy surprises. Put differently, it means that the spot exchange rate largely twists around an unchanged forward rate in response to monetary shocks. When a similar effect is observed in commodity markets – for example, when temporary supply shocks increase spot but not forward prices – the usual interpretation is that the shock is only expected to have a temporary effect on prices (Williams, 1986). If this interpretation is applied to the exchange rate market, it means that monetary policy shocks are only expected to have temporary effects on the exchange rate. This need not be the case; for example, according to conventional theory, a monetary loosening that was expected to lead to a long term increase in the inflation rate might be expected to lead to a decline in future spot rates and a decline in forward rates. This evidence that monetary surprises do not affect the long term forward exchange rate is consistent with other evidence that New Zealand financial markets believe the Bank's long term inflation goals will be achieved.

The second result concerns the effect of CPI and GDP surprises on financial market prices. In both cases, the paper shows that the spot exchange rate changes by nearly three times as much as five-year swap rates in response to news, appreciating when the CPI or GDP is higher than expected. Consequently, the change in the spot-forward exchange rate schedule is dominated by the shift in the long term forward rate rather than the twist in the schedule. Drawing a parallel with commodity markets again, it seems that CPI and GDP news largely have permanent rather than temporary effects on exchange rates.

¹⁰ Since the test associated with Eq. (17) has very poor properties, as $\hat{\alpha}_s/\hat{\alpha}_r > 5000$, we ignore it.

The third result just concerns the effect of CPI surprises on financial market prices. As stated above, the paper shows that the spot exchange rate changes by nearly three times as much as five-year swap rates in response to CPI news, appreciating when the CPI is higher than expected. It is difficult to make sense of this result in terms of traditional monetary models of the exchange rate, including models that include the endogenous response of monetary policy to news such as the forward-looking Taylor rule model of Engel et al. (2007). It means that New Zealand's forward exchange rates as well as the spot rate appreciate in response to bad news about inflation, a result that is not consistent with conventional economic models. This evidence is also in contrast with that of Faust et al. for the U.S. economy, where forward rates depreciate in response to bad news about inflation. Moreover, this evidence sits oddly with evidence from the monetary surprises regression that monetary policy largely has a temporary effect on the exchange rate.

One interpretation of this finding is that the spot exchange rate is excessively sensitive to CPI news, in the sense that it responds by much more than the response in interest rates. An alternative interpretation is based on the assumption that CPI news provides a signal of the strength of the underlying economy as well as a signal of the future direction of monetary policy. The exchange rate schedule largely shifts rather than twists in response to CPI news. This is similar to the response to GDP news, but in marked contrast to the response to monetary policy news. As such, it would appear that the exchange rate mainly responds to the GDP signal rather than to the monetary policy signal in CPI news. If this interpretation is correct, bad news about inflation is good news for the exchange rate not because of an anticipated monetary tightening but because the bad news about inflation reflects good news about the economy.

Overall, our findings are consistent with but extend the findings of the earlier literature that either analysed Taylor rule style models, such as Engel et al. (2007) or used an event-study approach, such as Clarida and Waldman (2007). To date, this literature has focused on the implications of news about fundamentals on the spot exchange rate. The models have implications for both the spot and forward exchange rates, however, implications that can be jointly analysed and tested. The simultaneous analysis of both the spot and forward exchange rates to news provides an avenue to examine whether the data are consistent with both a model of the fundamental long term determinants of exchange rates and the effect of interest rate differentials on short term exchange rate dynamics.

6. Conclusion

Recent analysis of the determinants of exchange rates has emphasized the way monetary policy responds endogenously to changing expectations of economic fundamentals such as inflation and output. This paper extends this analysis by examining the relative sensitivity of the spot exchange rate and interest rates to news. We show this is equivalent to examining the way the spot-forward exchange rate schedule responds to economic fundamentals, and further show that the different types of news should move the exchange rate schedule in different ways.

The paper has also suggested a new way of testing whether the spot exchange rate and the interest rate differential respond by the same amount to news. The alternative test takes advantage of potential information that may be observed by market participants but not observed by the econometrician. This information is likely to induce a correlation between the residuals of standard event analysis equations estimated separately, correlation that can be exploited to improve the efficiency of the tests in certain circumstances. The alternative test is simple to apply, although them improvements in efficiency in this sample appear to have been modest.

We have applied this technique to extend the New Zealand "even analysis" literature by comparing the relative size of the exchange rate and interest rate response to macroeconomic news. It has shown that spot exchange rates and five-year swap rate differentials respond by a similar amount in response to monetary policy surprises, but that the spot exchange rate changes by nearly three times as much as swap rates in response to CPI and GDP news. The CPI result in particular is intriguing, as it means that medium term forward rates as well as the spot rate appreciate in response to bad news about inflation. This evidence is not consistent with conventional economic models or empirical evidence about the relative size of exchange rate and interest rate differential response to CPI news in the United States. It may suggest that the New Zealand spot exchange rate is excessively responsive to CPI news,

or that CPI news largely provides a signal about the underlying strength of the economy. Discovering why New Zealand's spot exchange rate is so sensitive to CPI news would appear to warrant further research.

Acknowledgements

We would like to thank the seminar participants at the Reserve Bank of New Zealand, Econometric Society Australasia Meeting in 2008, and the anonymous referee.

Appendix A. The forward rate in a Taylor rule model

Consider the forward looking Taylor rule model considered by [Engel and West \(2006\)](#) and analysed in [Engel et al. \(2007\)](#). The home and foreign countries choose nominal interest rates according to the Taylor rules:

$$i_t = \gamma_q q_t + \gamma_\pi E_t \pi_{t+1} + \gamma_y y_t + \delta i_{t-1} + u_{mt} \quad (19)$$

$$i_t^* = \gamma_\pi E_t \pi_{t+1}^* + \gamma_y y_t^* + \delta i_{t-1}^* + u_{mt}^* \quad (20)$$

If the nominal exchange rates obey uncovered interest parity, subject to a risk premium $E_t s_{t+1} = s_t + (i_t - i_t^*) - \rho_t$ the nominal spot exchange rate equals

$$s_t = q_t + (p_t - p_t^*) = (p_t - p_t^*) + b \sum_{j=0}^{\infty} b^j E_t z_{t+j} \quad (21)$$

where

$$b = \frac{1}{1 + \gamma_q}$$

and

$$z_t = -[(\gamma_\pi - 1)E_t(\pi_{t+1} - \pi_{t+1}^*) + \gamma_y(y_t - y_t^*) + \delta(i_{t-1} - i_{t-1}^*) + (u_{mt} - u_{mt}^*) - \rho_t]$$

Define the 1 period forward interest rate for period $t+j$ contracted at time t as $i_{t,j}^1$, with $i_t \equiv i_{t,0}^1$. Let $v_{t,j}$ be the difference between forward rate and the expected future one period interest rate, $E_t i_{t+j}^1 = i_{t+j}^1 + v_{t,j}$ ($v_{t,0} = 0$). Define the n period interest rate at time t , i_t^n , as

$$i_t^n = \frac{1}{n} \sum_{j=0}^{n-1} i_{t,j}^1 = \frac{1}{n} \sum_{j=0}^{n-1} (E_t i_{t+j} - v_{t,j}) \quad (22)$$

The n period forward exchange rate at time t is

$$f_t^n = s_t + n(i_t^n - i_t^{n*}) = s_t + \sum_{j=0}^{n-1} (E_t i_{t+j} - v_{t,j}) - (E_t i_{t+j}^* - v_{t,j}^*) \quad (23)$$

The long term forward rate at time t is

$$f_t^\infty = s_t + n(i_t^\infty - i_t^{\infty*}) = s_t + \sum_{j=0}^{\infty} E_t (i_{t+j} - i_{t+j}^*) - (E_t i_{t+j}^* - v_{t,j}^*) \quad (24)$$

Now,

$$\begin{aligned}
 (i_{t+j} - i_{t+j}^*) &= \gamma_q q_{t+j} + \gamma_\pi E_{t+j}(\pi_{t+j+1} - \pi_{t+j+1}^*) + \gamma_y (y_{t+j} - y_{t+j}^*) \\
 &\quad + \delta(i_{t+j-1} - i_{t+j-1}^*) + (u_{mt+j} - u_{mt+j}^*) \\
 &= \gamma_q q_{t+j} - z_{t+j} + \rho_{t+j} + E_{t+j}(\pi_{t+j+1} - \pi_{t+j+1}^*) \\
 &= \gamma_q b \sum_{k=0}^{\infty} b^k E_{t+j} z_{t+j+k} - z_{t+j} + \rho_{t+j} + E_{t+j}(\pi_{t+j+1} - \pi_{t+j+1}^*)
 \end{aligned} \tag{25}$$

Applying the law of iterated expectations, $E_t[E_{t+j}[X_\tau]] = E_t[X_\tau]$, and noting $\gamma_q b/(1-b) = 1$

$$\begin{aligned}
 \sum_{j=0}^{\infty} E_t(i_{t+j} - i_{t+j}^*) &= \gamma_q b \sum_{k=0}^{\infty} b^k E_t z_{t+k} - z_t + \rho_t + E_t(\pi_{t+1} - \pi_{t+1}^*) \\
 &\quad + \gamma_q b \sum_{k=0}^{\infty} b^k E_t z_{t+1+k} - E_t z_{t+1} + E_t \rho_{t+1} + E_t(\pi_{t+2} - \pi_{t+2}^*) \\
 &\quad + \gamma_q b \sum_{k=0}^{\infty} b^k E_t z_{t+2+k} - E_t z_{t+2} + E_t \rho_{t+2} + E_t(\pi_{t+3} - \pi_{t+3}^*) \dots \\
 &= (\gamma_q b - 1) z_t + \rho_t + E_t(\pi_{t+1} - \pi_{t+1}^*) + [\gamma_q b(1+b) - 1] E_t z_{t+1} \\
 &\quad + E_t \rho_{t+1} + E_t(\pi_{t+2} - \pi_{t+2}^*) \\
 &\quad + [\gamma_q b(1+b+b^2) - 1] E_t z_{t+2} + E_t \rho_{t+2} + E_t(\pi_{t+3} - \pi_{t+3}^*) \dots \\
 &= \left[\frac{\gamma_q b}{1-b} (1-b) - 1 \right] z_t + \rho_t + E_t(\pi_{t+1} - \pi_{t+1}^*) \\
 &\quad + \left[\frac{\gamma_q b}{1-b} (1-b^2) - 1 \right] E_t z_{t+1} + E_t \rho_{t+1} + E_t(\pi_{t+2} - \pi_{t+2}^*) \\
 &\quad + \left[\frac{\gamma_q b}{1-b} (1-b^3) - 1 \right] E_t z_{t+2} + E_t \rho_{t+2} + E_t(\pi_{t+3} - \pi_{t+3}^*) \dots
 \end{aligned} \tag{26}$$

which is equal to the following:

$$-b \sum_{j=0}^{\infty} b^j E_t z_{t+j} + \sum_{j=0}^{\infty} E_t \rho_{t+j} + \sum_{j=0}^{\infty} E_t(\pi_{t+j+1} - \pi_{t+j+1}^*) \tag{27}$$

Hence

$$\begin{aligned}
 f_t^\infty &= s_t + \sum_{j=0}^{\infty} E_t(i_{t+j} - i_{t+j}^*) - (v_{t,j} - v_{t,j}^*) \\
 &= (p_t - p_t^*) + b \sum_{j=0}^{\infty} E_t b^j E_t z_{t+j} - b \sum_{j=0}^{\infty} E_t b^j E_t z_{t+j} + \\
 &\quad + \sum_{j=0}^{\infty} E_t \rho_{t+j} + \sum_{j=0}^{\infty} E_t(\pi_{t+j+1} - \pi_{t+j+1}^*) - \sum_{j=0}^{\infty} (v_{t,j} - v_{t,j}^*) \\
 &= (p_t - p_t^*) + \sum_{j=0}^{\infty} E_t(\pi_{t+j+1} - \pi_{t+j+1}^*) - \sum_{j=0}^{\infty} (v_{t,j} - v_{t,j}^*)
 \end{aligned} \tag{28}$$

The first two terms represent the long term expected future spot rate, which is equal to the ratio of the prices in each country. The third and fourth terms are the sum of the risk parameters: $E_t \rho_{t+j}$ is the exchange rate risk premium, equal to the difference between the expected 1 period forward rate at time $t+j$ and the expected future spot rate at $t+j+1$; and $v_{t,j}$ is the difference between the 1 period forward interest rate $i_{t,j}^1$ and the expected future 1 period interest rate $E_t i_{t+j}$. Together they are equal to the difference between the forward exchange rate and the expected future exchange rate.

Consider an event occurring at time t , let $E_{t,-}$ and $E_{t,+}$ be the expectations immediately before and after the event, and $\Delta E_t[X] = E_{t,+}[X] - E_{t,-}[X]$. Then

$$\Delta E_t[f_t^\infty] = \left[\Delta E_t(p_t - p_t^*) + \sum_{j=0}^{\infty} \Delta E_t(\pi_{t+j+1} - \pi_{t+j+1}^*) + \sum_{j=0}^{\infty} \Delta E_t \rho_{t+j} - \sum_{j=0}^{\infty} \Delta E_t(v_{t,j} - v_{t,j}^*) \right] \quad (29)$$

Table 6

NZ–US dollar response to news.

Dependent variable	Window	W	$\hat{u}_r$	$\Delta \hat{R}$	Obs	R^2	t-Test $\alpha_s = \alpha_r$
5-Year rate response to OCR surprises							
Δf_5	Tight	0.0098 (0.011)			54	0.02	0.92
Δf_5	Wide	0.012 (0.0084)			55	0.04	1.44
Δf_5	Daily	0.011 (0.010)			55	0.02	1.05
Δs	Tight		0.87** (0.30)	1.45** (0.50)	54	0.25	0.89
Δs	Wide		1.06** (0.33)	1.51** (0.36)	55	0.35	1.41
Δs	Daily		0.87** (0.30)	1.49** (0.48)	55	0.26	1.01
5-Year rate response to CPI surprises							
Δf_5	Tight	0.0055* (0.0022)			23	0.23	2.49*
Δf_5	Wide	0.049* (0.0020)			23	0.18	2.13*
Δf_5	Daily	0.0063 (0.0043)			23	0.09	1.48
Δs	Tight		2.41** (0.78)	2.71** (0.66)	23	0.57	2.61*
Δs	Wide		0.57 (0.50)	2.27** (0.76)	23	0.44	1.68
Δs	Daily		0.17 (0.53)	1.91* (0.74)	23	0.34	1.23
5-Year rate response to GDP surprises							
Δf_5	Tight	0.0042* (0.0016)			23	0.24	2.54*
Δf_5	Wide	0.040 (0.0021)			23	0.14	1.87
Δf_5	Daily	0.0029 (0.0013)			23	0.04	0.88
Δs	Tight		2.35** (0.59)	2.89** (0.68)	23	0.63	2.73*
Δs	Wide		2.42** (0.48)	2.50** (0.69)	23	0.66	2.18*
Δs	Daily		1.15 (0.71)	2.04 (1.26)	23	0.22	0.83

* Statistical significance at 5% level.

** Statistical significance at 1% level.

This equation says that the forward rate should appreciate or depreciate (a) by the amount the event changes perceptions about the long term change in the expected spot rate, equal to the ratio of prices in each country and (b) by the extent that the event changes risk premia. Assuming the event does not change the risk premia, therefore, the long term forward rate should depreciate if domestic prices are expected to increase, and appreciate if domestic prices are expected to decrease. It follows that even if an increase in inflation makes the spot rate appreciate, it should make the long term forward rate depreciate. In contrast, while an increase in GDP should make the spot rate appreciate, it should only lead to an appreciation of the forward exchange rate if local prices are expected to fall in the long run.

Appendix B. Regression results with different period windows

As discussed in Section 3, we calculate the response of interest rate and exchange rate changes to news announcements over three different periods: the one hour tight window, the two hour wide window, and the one day daily window. Our preference is to use the tight window data, although we use the wide window data for the monetary policy surprises as the tight window ends only 10 min after the announcement. We prefer the short windows because there is less scope for the exchange rates or interest rates to change in response to other factors during short periods of time, and the variance of the term $u_t^{i,j}$ in the regression

$$\Delta P_t^i = \alpha_0^{i,j} + \alpha_1^{i,j} W_t^j + u_t^{i,j} \quad (30)$$

should be smaller.

Table 6 shows the results of the main equations presented in Tables 2–5 estimated using different length windows. In general, the coefficient estimates for each equation are similar for the different windows, but the t -test of the hypothesis that $\alpha_s = \alpha_r$ get smaller as the window period is increased. In particular, using the tight window it is possible to reject the hypothesis that $\alpha_s = \alpha_r$ for the CPI and GDP data; using the daily window it is not.

The main case where there is a change in the coefficient estimates as the period of the window increases concerns the estimate of θ_1 in Eq. (17) for CPI surprises:

$$\Delta s = \Delta \hat{R}^* \theta_0 + \hat{u}_r \theta_1 + \xi \quad (31)$$

this coefficient is estimated to be 2.41 (s.e. = 0.78) in the tight window, but 0.57 in the wide window and 0.17 in the daily window. This coefficient is the same as the coefficient δ in Eq. (12):

$$\hat{u}_s = \delta \hat{u}_r + \eta \quad (32)$$

and thus measures the correlation between the residual terms in the basic event analysis regressions. Further analysis of this equation shows there is very high correlation between the residuals $\hat{u}_s$ and $\hat{u}_r$ in the first 25 min after the news announcement, but that in the next hour the innovations to interest rates and the spot exchange rate were uncorrelated. In addition, there was weak evidence of negative serial correlation between the change in the exchange rate in the first 25 min and the change in the exchange rate in the subsequent hour, although this effect was not large. It would appear, therefore that by moving from a tight window to a wide window there is considerable dilution of the underlying correlation between the residuals that is directly related to the announcement effect.

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